# Critical Analysis

# On *Variational Tensor Neural Networks for Deep Learning*

by Saeed S. Jahromi and Román Orús — *Scientific Reports*, August 2024

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## 1 Introduction and State of the Art

The article analysed here, *Variational Tensor Neural Networks for Deep Learning*, was first posted on arXiv in November 2022 and published in *Scientific Reports* in August 2024 by Saeed S. Jahromi and Román Orús [1], with co-authorship from Multiverse Computing, a quantum-inspired software company.

The authors frame their work as a response to two long-standing limitations of deep learning: the explosion of parameters in large networks, and the *opacity* of trained weights, which reveal little about what a model has actually learned. Their solution is a Tensor Neural Network (TNN) trained with an algorithm inspired by DMRG (Density Matrix Renormalization Group, originally introduced by White [2]).

To assess the genuine novelty of the paper, it must be placed inside an already mature research line. On the side of *compression*, Novikov *et al.* [3] introduced "tensorizing neural networks" in 2015: fully-connected layers stored in tensor-train format, trained by ordinary back-propagation and inserted among standard convolutional and dense layers, with compression factors of up to four orders of magnitude and negligible accuracy loss. Gao *et al.* [4] extended this idea by compressing entire deep networks with Matrix Product Operators (MPOs); Patel *et al.* [5] used MPO-based TNNs to solve partial differential equations; Wang *et al.* [6] deployed tensor networks for anomaly detection. On the side of *training*, Stoudenmire and Schwab [7] showed that a single Matrix Product State (MPS), used as the whole model, could perform supervised classification when trained with a DMRG-like sweep, and Han *et al.* [8] trained MPS generative models with the same kind of two-site updates. On the side of *interpretation*, Cohen, Sharir and Shashua [9] recast deep networks as hierarchical tensor decompositions; Levine *et al.* [10] then used quantum entanglement measures to quantify the correlations a deep convolutional network can model; and the entanglement of trained MPS classifiers has been used to rank input features [11] and to study what image classification requires [12]. On the physics side, the survey of Verstraete *et al.* [13] and the practical introduction by Orús [14] had long established the variational principles shared by tensor networks and quantum many-body states. Tensorised layers, hybrid stacks containing them, DMRG-style training and entanglement-based readings of tensor-network models therefore all predate this paper.

What the paper contributes is the *junction* of these lines. It trains MPO layers *embedded in a deep, hybrid neural network*, freely mixed with dense layers, with a variational two-site sweep borrowed from DMRG, where earlier hybrid networks used back-propagation and earlier DMRG-style training applied only to standalone tensor-network models. As a consequence, the entanglement spectrum of the weights becomes an observable of the network *during training*. The paper brings no major advance in compression, which was already established [3, 4], nor in scalability, which its own timings contradict

(Section 3). Its contribution is a new dimension of *interpretability* — understood here as the ability to attach a quantitative, humanly meaningful reading to a model's internal state, rather than merely observing that it maps inputs to outputs correctly — obtained through a new combination of training procedure and network structure. This is the angle on which the present analysis will focus.

## 2 Concept and Mathematical Formulation

A tensor network factorises a large weight matrix into a chain of small connected tensors. Originating in quantum physics for representing many-body wave functions, it doubles as a compression scheme in deep learning: only the components carrying meaningful correlations are retained, and the rest discarded.

The authors use the **Matrix Product Operator** (MPO). A weight matrix $M \in \mathbb{R}^{p^2 \times p^2}$ is reshaped to rank 4 and decomposed via the Singular Value Decomposition (SVD):

$$M = USV^\dagger = LR, \qquad L = U\sqrt{S}, \qquad R = \sqrt{S}\,V^\dagger. \tag{1}$$

The diagonal matrix $S$ collects the singular values $\{\lambda_i\}$, and truncating to the $\chi_t \leq \chi$ largest of them defines the *bond dimension*. This parameter controls how much information passes between tensors — and, crucially, it quantifies the *entanglement* (correlation) between different parts of the network.

The full tensorisation procedure is summarised in Fig. 1. Panel (a) shows the classic dense layer $y_p = \sigma(Wx + b)$; panel (b) shows how the $216 \times 216$ weight matrix $W$ is reshaped and split, via successive SVDs, into three smaller MPO tensors $w_1, w_2, w_3$ linked by virtual bonds of dimension $\chi$; panel (c) shows the resulting TNN forward pass. The key visual insight is that the wide blue lines of width 216 are replaced by three narrow red legs of width 6 each, with controlled black bonds $\chi$ in between. This structural change is what compresses the layer; as shown below, it is the way the layer is *trained* that turns the bonds into a window on what has been learned.

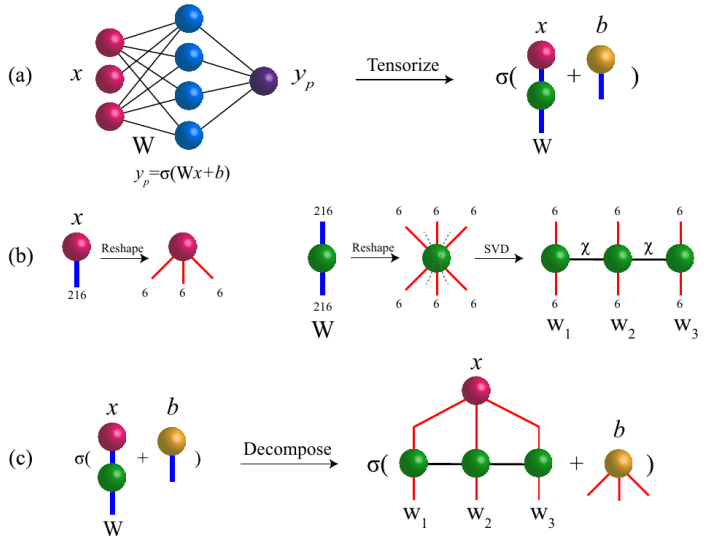



**Figure 1.** Tensorisation of a dense layer into a TNN (reproduced from Fig. 5 of [1]). (a) Standard NN → tensor diagram. (b) SVD-based decomposition of the weight matrix into three MPO tensors. (c) Resulting TNN forward pass.

The forward pass replaces the usual $y_p = \sigma(Wx + b)$ by the contracted form

$$y_p = \sigma(\mathrm{tTr}[x_{ijk\ldots}\, w_i w_j w_k \ldots] + b_{ijk\ldots}), \tag{2}$$

where tTr denotes the tensor trace over virtual indices.

For training, the authors adapt DMRG — a standard tool in computational quantum physics. Instead of optimizing all weights globally as Adam or SGD would, the algorithm sweeps through tensor *pairs*, contracts each pair $w_i, w_{i+1}$ into a *bond tensor* $B_{i,i+1}$, and updates it by a local gradient step. For the mean-squared-error loss $L_{\mathrm{MSE}} = \frac{1}{N_s}\sum_{n=1}^{N_s}(y_p^{(n)} - y^{(n)})^2$ over $N_s$ training samples, with $y_p = E\,B$,

$$\frac{\partial L_{\mathrm{MSE}}}{\partial B} = -\frac{2}{N_s}\sum_{n=1}^{N_s}\left(y^{(n)} - y_p^{(n)}\right)E^{(n)}, \qquad B'_{i,i+1} = B_{i,i+1} - \alpha\,\frac{\partial L_{\mathrm{MSE}}}{\partial B_{i,i+1}}, \tag{3}$$

where $E$ is the contraction of the whole network except $B$ (the environment of the bond tensor, written $M$ in [1])[1]. For non-linear activations the same gradient is obtained by automatic differentiation. The updated bond tensor is re-split by SVD, $B' = U'S'V'^{\dagger}$, and truncated — which both controls the parameter count and *exposes the singular values at every step*. This is the key technical move: with the two-site update, the spectrum is a by-product of training itself. Training all tensor cores at once by back-propagation, as in the tensorised networks of Novikov *et al.* [3], never forms this bond tensor and leaves the spectrum hidden unless it is computed separately. The two-site sweep is not new in itself — it is the heart of DMRG, and was already used for standalone MPS models [7, 8] — but here it acts on MPO layers that are part of a neural network.

The internal correlations of the network are then summarised by the **entanglement entropy**

$$S = -\sum_i \lambda_i^2 \log \lambda_i^2, \qquad \textstyle\sum_i \lambda_i^2 = 1, \tag{4}$$

computed at every virtual bond from the normalised singular values (a layer of $N_{\mathrm{MPO}}$ tensors has $N_{\mathrm{MPO}} - 1$ bonds; the authors report the middle one). A product-state-like model ($S \approx 0$) signals weak parameter correlation; large $S$ signals a strongly correlated, expressive representation. This is precisely where the interpretability gain becomes concrete. Where a conventional network exposes only a scalar loss — a measure of how well it fits, but mute on *what* it has internally organised — the entropy $S$ is a number read directly off the trained weights that characterises the correlation structure they encode. The opacity described in the introduction is thus not lifted by an external probing tool applied after training, but by a quantity intrinsic to the architecture and available throughout training. In short, two-site training of MPO layers turns **correlation into a measurable quantity**.

The second ingredient is the **hybrid structure**. Because the gradient of each bond tensor can be obtained by automatic differentiation, `TNLayer` blocks are stacked with ordinary `Dense` layers in a single model and a single training loop, the dense weights being updated by plain gradient descent. Tensor layers inside ordinary networks already existed [3, 4]; what is new is to train them variationally, so that the entanglement reading of tensor-network models becomes available within an ordinary deep-learning pipeline.

## 3 Validation

The authors benchmark their method across four families of tasks — linear and non-linear regression, classification of Gaussian blobs and spirals, and MNIST image recognition. Table 1 collects the quantitative outcomes; the most diagnostic of them — and the cleanest evidence for the interpretability claim — come from the two-dimensional classifiers.

[1] Eq. (7) of [1] defines $\Delta B = -\partial L_{\mathrm{MSE}}/\partial B$ and updates $B' = B - \alpha\,\Delta B$, which, taken literally, is a gradient *ascent*; it also sums over $n$ a term indexed by $j$. The consistent form is given here.

| **Task** | **Architecture** | **Accuracy / Fit** | **Entropy** $S$ |
|---|---|---|---|
| Linear regression | 1 TNLayer ($N_{\mathrm{MPO}} = 6$) | $a_p = 0.57,\ b_p = 0.19$ | — |
| Non-linear regression | 3 TNLayers (Linear/Sigmoid) | near-perfect $\sin x$ fit | — |
| Gaussian blobs | 1 TNLayer + 2 Dense | 99% | $\approx 0$ |
| Spiral distribution | 1 TNLayer + 2 Dense | 95% | $\approx 0.7$–$0.8$ |
| MNIST (TNN, 1 epoch) | 1 TNLayer ($N_{\mathrm{MPO}} = 12$) | 95.8% | — |
| MNIST (classical, 1 epoch) | Dense baseline (Adam) | 94.2% | — |
| MNIST (classical, 60 epochs) | Dense baseline (Adam) | 97.6% | — |

**Table 1.** Summary of the benchmark results reported in [1]. For the linear regression, $a_p$ and $b_p$ are the predicted slope and intercept, against true values $a = 0.55$ and $b = 0.20$. The entropy column is reported only for the classification tasks, where the entanglement structure of the trained weights was monitored.

The regression and blob tasks confirm that the TNN fits standard data accurately, but they are not where the contribution lies. The decisive comparison is between the two classifiers, both hybrid networks (Table 1). On the **Gaussian blobs**, the model reaches 99% accuracy and the entanglement entropy collapses to nearly zero across all bond dimensions $\chi$, faithfully reflecting the clean separation of the two clusters. On the **spiral distribution**, the picture changes qualitatively. Fig. 2 shows the trained decision boundary winding cleanly through three interlocking arms, with an overall accuracy near 95%. What matters for the interpretability claim, however, is not this accuracy but the behaviour of the entanglement entropy reported by the authors: while the loss drops monotonically for all bond dimensions $\chi \in \{6, 10, 16\}$, the entropy $S$ does *not* decay to zero. Instead it converges to a non-trivial plateau around $S \approx 0.7$–$0.8$. This non-vanishing entropy is the algebraic signature that the trained MPO weights are genuinely entangled — precisely the situation one expects when the input data are geometrically intertwined and cannot be separated by any product-state model. The internal entanglement measure thus *mirrors the geometric structure of the data* — well-separated data $\rightarrow$ low $S$; intertwined data $\rightarrow$ high $S$, as the entropy column of Table 1 makes explicit. Hence, reading a single scalar off the trained network, while it trains, already gives actual insight into the data it has learned, which a standard dense network could not deliver.

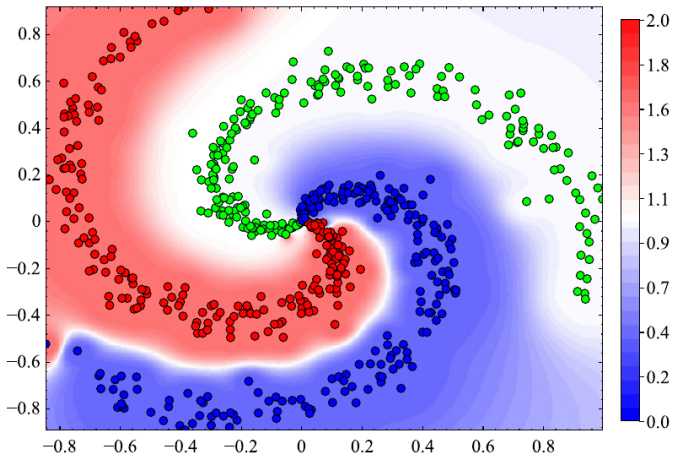



**Figure 2.** Ternary classification of a random spiral distribution. Filled circles are training data; red, white, blue shaded regions are the decision boundary obtained by the TNN. From Fig. 13 of [1].

On **MNIST**, the TNN reaches 95.8% test accuracy after a single training epoch at $N_{\mathrm{MPO}} = 12$, marginally above the 94.2% obtained by a classical fully-connected baseline trained for one epoch with Adam (Table 1). This single-epoch comparison is fragile: the same classical baseline climbs to 97.6% once trained to convergence over 60 epochs, and the standalone MPS classifier of Stoudenmire and Schwab [7], which the authors plot for reference, had already reported a test error below 1% at large bond dimension. More damningly for the scalability claim, the authors themselves report that the training time per epoch of the DMRG-like algorithm grows *exponentially* with the number of MPO tensors $N_{\mathrm{MPO}}$, reaching minutes against roughly ten seconds for the Adam-trained baseline. Nor are the parameter counts compared with those of earlier tensorised layers [3, 4], so no claim of better compression can be drawn either. The MNIST result is therefore best read as a proof of concept — which the authors openly acknowledge — rather than a claim of competitive performance.

## 4 Conclusion

Stripped of the abstract's framing, the paper of Jahromi and Orús is not a breakthrough in compression — tensorised layers had already shown it [3, 4] — nor in scalability, which its own timings contradict. Neither are entanglement-based readings of learned tensor-network models new [10, 11, 12]. Its genuine contribution is a new *combination* of structure and training: MPO layers embedded in hybrid deep networks alongside dense layers, and trained by two-site variational sweeps rather than by back-propagation. This combination makes the entanglement spectrum a first-class observable of a neural network *during training*. Through the singular values exposed at each sweep and the entropy $S = -\sum_i \lambda_i^2 \log \lambda_i^2$, the network's internal correlation structure becomes a directly measurable quantity, aligned with the geometric complexity of the data, as Fig. 2 and Table 1 show. This is the precise sense in which the work opens a new dimension of interpretability: it replaces the opaque, loss-only view of a training network with an intrinsic, quantitative reading of what the model has organised internally. In an era dominated by large language models whose internal representations remain almost entirely opaque, such a reading offers interpretability research a rare quantitative anchor — rooted in the well-developed mathematical language of quantum information — provided the training cost can be brought under control.

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